

Assignment 10.2 - Sequences

1. Determine if the following sequences converge or diverge. If it converges, find  $L$ .

a.  $\left\{ \frac{3n^3-1}{2n^3+1} \right\} \lim_{n \rightarrow \infty} \left( \frac{3n^3-1}{2n^3+1} \right) = \lim_{n \rightarrow \infty} \left( \frac{\frac{3n^3}{n^3} - \frac{1}{n^3}}{\frac{2n^3}{n^3} + \frac{1}{n^3}} \right) = \lim_{n \rightarrow \infty} \left( \frac{3 - \frac{1}{n^3}}{2 + \frac{1}{n^3}} \right) = \frac{3-0}{2+0} = \frac{3}{2}$

Converge:  $L = \frac{3}{2}$

b.  $\left\{ \arctan \frac{10n}{10n+4} \right\} \rightarrow \lim_{n \rightarrow \infty} \left( \arctan \left( \frac{10n}{10n+4} \right) \right) = \arctan \left( \lim_{n \rightarrow \infty} \frac{10n}{10n+4} \right) = \arctan \left( \lim_{n \rightarrow \infty} \frac{10}{10+\frac{4}{n}} \right) = \arctan(1) = \frac{\pi}{4}$

Converge:  $L = \frac{\pi}{4}$

c.  $\left\{ \frac{\ln n^2}{\ln 3n} \right\} \rightarrow \lim_{n \rightarrow \infty} \frac{\ln n^2}{\ln 3n} = \lim_{n \rightarrow \infty} \left( \frac{2 \ln n}{\ln 3 + \ln n} \right) = \lim_{n \rightarrow \infty} \left( \frac{2}{\frac{\ln 3}{\ln n} + 1} \right) = \frac{2}{0+1} = 2$

Converge:  $L = 2$

d.  $\left\{ \frac{(n+1)!}{(n)!} \right\} = \frac{(n+1)(n)!}{n!} \rightarrow \lim_{n \rightarrow \infty} \left( \frac{(n+1)(n)!}{(n)!} \right) = \lim_{n \rightarrow \infty} (n+1) = \infty$  diverges

e.  $\left\{ \sqrt[n]{e^{3n+4}} \right\} \lim_{n \rightarrow \infty} \left( e^{3n+4} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left( e^{3+\frac{4}{n}} \right) = e^3$  converges  $L = e^3$

2. Find the limit of the following sequences or state that they diverge. Make sure to state which sequence grows faster by the Growth Rate Theorem, if it applies.

a.  $\left\{ \frac{\ln n^8}{0.001n} \right\}$   $\ln n^8 \rightarrow 8 \ln n$   $\log \ll \text{power}$   
 $0.001n \rightarrow 0.001n$

So by GRT,  $n$  grows faster than  $\ln n$  so it converges to 0.  $L = 0$

b.  $\left\{ \frac{n!}{100^n} \right\}$   $n! \rightarrow \text{factorial}$   $\text{expo} \ll \text{factorial}$   
 $100^n \rightarrow \text{exponential}$

So by GRT,  $n!$  grows faster than  $100^n$  so it diverges to  $\infty$ .

c.  $\left\{ \frac{3^{5n}}{n!} \right\}$   $3^{5n} = (3^5)^n$   
 $n! \ll \text{factorial}$

So by GRT,  $n!$  grows faster than  $243^n$  so it converges to 0.  $L = 0$

3. Find the limit of the sequence  $b_n = \frac{\cos n}{n^2+1}$  (SQZTHM).

$\frac{-1}{n^2+1} \leq \frac{\cos n}{n^2+1} \leq \frac{1}{n^2+1}$   $\lim_{n \rightarrow \infty} \left( \frac{-1}{n^2+1} \right) = 0$   $\lim_{n \rightarrow \infty} \left( \frac{1}{n^2+1} \right) = 0$   $b_n = \frac{\cos n}{n^2+1}$  as  $n$  approaches infinity is 0.

4. Explain what it means to converge monotonically versus by oscillation. Give an example.

monotonically  $\rightarrow$  Sequence approaches the limit by always moving in one direction, increasing or decreasing

oscillation  $\rightarrow$  Sequence swings or jumps back and forth/above and below the limit but gets smaller and smaller until settles on the limit

5. Explain what it means for a sequence to be bounded. Give an example.

A sequence is bounded if there is a ceiling and a floor that it never goes past